



CHALMERS
UNIVERSITY OF TECHNOLOGY

MTF236 - Road Vehicle Aerodynamics
VT2025 Lp3 - Caster Lab Assignment

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Question 1

Without aero package

1. The steering felt lighter and more responsive at lower speed, but at higher speed the vehicle exhibited instability during cornering.
2. Due to lack of aerodynamic downforce there was reduced traction due to which the vehicle exhibits understeer or oversteer during cornering.
3. At speed around 130 km/hr during cornering due to reduced traction and downforce it was difficult to keep the vehicle on the track without skidding out.

With Aero Package

1. The vehicle with aero package provided additional downforce due to which high speed stability and cornering grip were improved. However, this also increased the weight of the steering.
2. The vehicle was more predictable compared to the one without the aero package, and it reduced sudden understeer or oversteer during cornering.
3. Even at high speeds of around 155 km/hr during cornering, the vehicle was stable with good traction, and it was easy to keep the vehicle on the track.

Question 2

The lateral acceleration curves for the vehicle both with and without the aerodynamic package were plotted from the smoothened raw data and the intervals of time where the acceleration values were relatively constant were selected as shown in Figure 1. The velocities at those times were noted down as well (all values listed in Table 1).

It was possible to take curves faster, and accelerate harder sideways, with the aero-package than without. In both cases, it was possible to take the larger curve at a greater speed than the smaller curve, as expected. The lateral acceleration was also higher on the larger curve in both cases.

<u>Type</u>	<u>Time (s)</u>	<u>Lat. acc (m/s^2)</u>	<u>Forw. vel. (m/s)</u>
<i>Without aero</i>			
Large corner	79	13.23	33.30
Small corner	175	-13.02	32.00
<i>With aero</i>			
Large corner	133	18.5	40.00
Small corner	110	-16.5	33.10

Table 1: Lateral accelerations and linear velocities in selected regions



Figure 1: Real-time data of vehicle lateral acceleration and velocity, showing the selected steady-state regions for the no-aero test marked in yellow, and those for the test with aero marked in green.

Question 3

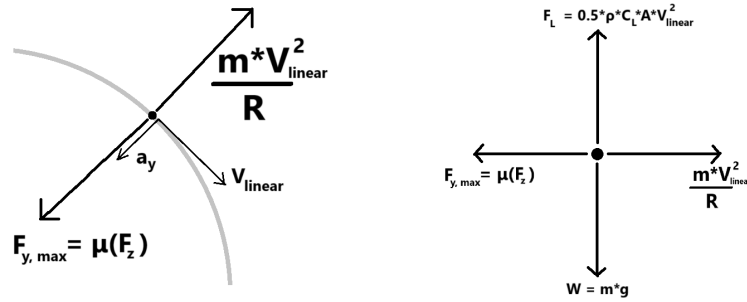


Figure 2: Free body diagram of a point-mass vehicle taking a curve of radius R (left: top view, right: front view)

Solving from the right-side FBD for a vehicle of $-C_L A = 4m^2$,

$$F_z = W - F_L$$

Thus,

$$\mu(W - F_L) = \frac{mV_{\text{linear}}^2}{R}$$

$$\mu \left(mg - \frac{1}{2} \rho C_L A V_{\text{linear}}^2 \right) = \frac{mV_{\text{linear}}^2}{R}$$

$$\mu mg = (V_{\text{linear}}^2) \left[\frac{m}{R} + \frac{\mu \rho C_L A}{2} \right] \dots\dots\dots(3.1)$$

$$[1.5 \times 1300 \times 9.81] = (V_{\text{linear}}^2) \left[\frac{1300}{103} + \frac{1.5(1.205)(-4)}{2} \right]$$

$$\Rightarrow V_{\text{linear}} = 46.086 \text{ m/s}$$

$$a_y = \frac{V_{\text{linear}}^2}{R} = \frac{46.086^2}{103} = 20.621 \text{ m/s}^2$$

And for a lift-neutral vehicle ($-C_L A = 0$),

$$\mu mg = V_{\text{linear}}^2 \left(\frac{m}{R} \right)$$

$$\Rightarrow V_{\text{linear}} = \sqrt{\mu g R} \\ = \sqrt{1.5 \times 9.81 \times 103} = 38.93 \text{ m/s}$$

$$\Rightarrow a_y = \frac{V_{\text{linear}}^2}{R} = \frac{38.93^2}{103} = 14.7140 \text{ m/s}^2$$

The steady-state velocity and acceleration from the Caster test with the aero switched on is found to be greater than the corresponding theoretical values for the lift-neutral vehicle, but lower than those calculated for the theoretical best performing vehicle with $-C_L A = 4$. This is because an increase in $-C_L A$ means an increase in downforce, which increases traction and allows the driver to take turns at greater speed.

However, it was not possible to attain the theoretical best performance due to lack of driver skill in keeping a constant turn radius and constant speed. The performance of the Caster vehicle without the aero package is lower than the theoretical $-C_L A = 0$ vehicle for the same reason.

Question 4

The equations of μ , m , and V_{linear} in terms of $C_L A$ are as below, derived from Equation 3.1. The known values are substituted in each case to generate the simplified equations of the graphs in Figure 3, which show how each of these three properties vary with $C_L A$ when everything else is kept constant.

$$V_{\text{linear}} = \sqrt{\frac{2R\mu mg}{2m + \mu \rho C_L A R}} = \sqrt{\frac{2 \times 103 \times 1.5 \times 1300 \times 9.81}{2600 + 1.5 \times 1.205 \times 103 \times C_L A}} \\ \Rightarrow V_{\text{linear}} = \frac{1985.11}{\sqrt{2600 + 186.17 \times C_L A}} \Rightarrow V_{\text{linear}} = \frac{145.49}{\sqrt{13.97 + C_L A}}$$

$$m = \frac{\mu \rho R C_L A V_{\text{linear}}^2}{2 \times (\mu g R - V_{\text{linear}}^2)} = \frac{1.5 \times 1.205 \times 103 \times C_L A \times 46.086^2}{2 \times (1.5 \times 9.81 \times 103 - 46.086^2)} \Rightarrow m = -325.03 C_L A$$

$$\mu = \frac{2mV_{\text{linear}}^2}{R(2mg - \rho C_L A V_{\text{linear}}^2)} = \frac{2 \times 1300 \times 46.086^2}{103 \times [(2 \times 1300 \times 9.81) - (1.205 \times 46.086^2 \times C_L A)]}$$

$$\Rightarrow \mu = \frac{53613.5}{25506 - 2559.32 C_L A} \Rightarrow \mu = \frac{20.95}{9.966 - C_L A}$$

The frictional force is given by $F_y = \mu F_z$ where μ is the friction coefficient and F_z is the normal force. And from the free-body diagram in Question 3, it is clear that increasing the $-C_L A$ increases F_z by reducing lift (or, in other words, increasing downforce).

- For the same mass and velocity, increasing the downforce F_z allows the friction coefficient to be lower for the same F_y , as shown in the plot on the left in Figure 3.
- For the same friction coefficient and velocity, increasing the downforce and thus increasing the frictional force F_y that the road can offer allows the road to resist a greater lateral force from the tyres, which means that the car can face a greater centrifugal force. This allows the maximum mass of the vehicle to be higher, as shown in the plot in the middle in Figure 3.

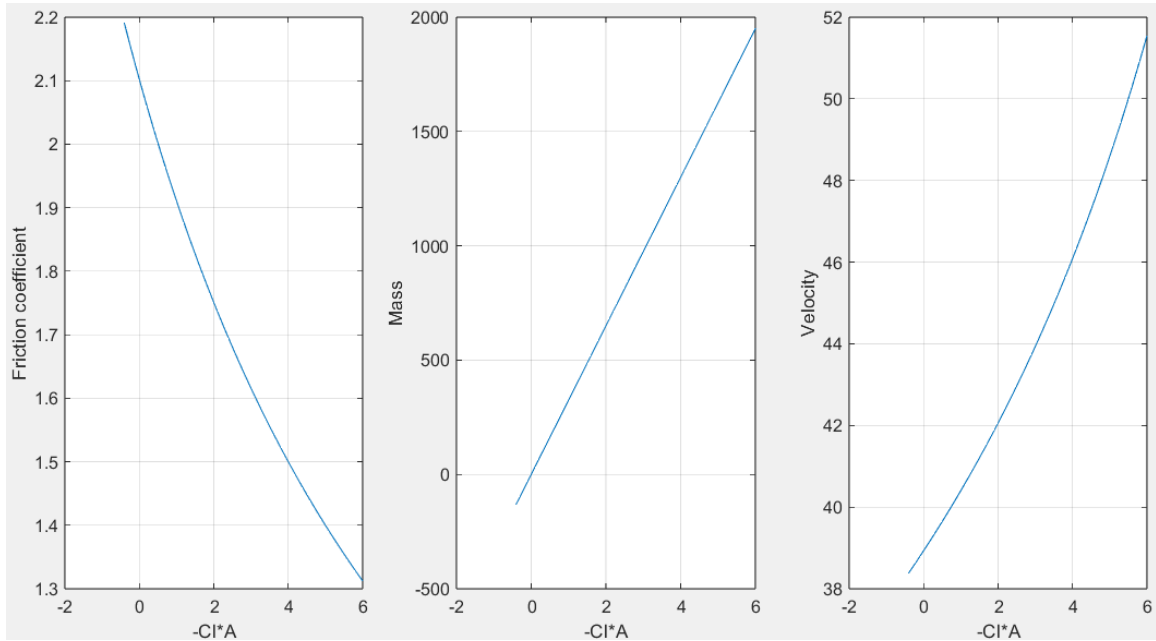


Figure 3: Plots of $-C_L A$ versus friction coefficient, mass, and velocity

Tradeoffs: For passenger vehicle design, while increasing $-C_L A$ (i.e., increasing downforce) it is preferable to keep the desired velocity and friction coefficient for a curve constant, while allowing the vehicle's mass to increase, to account for an increase in the number of passengers within the vehicle while still designing the vehicle to meet the same cornering performance.

Question 5

From the mass- $C_L A$ relation calculated in Question 4,

$$m = -325.03 C_L A$$

$$-C_L A = \frac{1300 + 7}{-325.03} = 4.0212 m^2$$

Thus, an increase of $7kg$ in mass necessitates an increase of $4.0212 - 4 = 0.0212m^2$ in $-C_L A$ to ensure that the vehicle has the same performance *i.e.* the same V_{linear} and a_y taking a curve of the same radius on a surface with the same friction coefficient.

Question 6

For a vehicle with $-C_L A = 4m^2$, the forward velocity varies with curve radius as below, starting from the first relation calculated in Question 4.

$$V_{\text{linear}} = \sqrt{\frac{2R\mu mg}{2m + \mu\rho C_L A R}} = \sqrt{\frac{R \times 2 \times 1.5 \times 1300 \times 9.81}{2600 + 1.5 \times 1.205 \times (-4) \times R}} \Rightarrow V_{\text{linear}} = \frac{195.6\sqrt{R}}{\sqrt{2600 - 7.23R}}$$

Similarly, for a vehicle with $-C_L A = -0.4m^2$, the relation is:

$$V_{\text{linear}} = \sqrt{\frac{2R\mu mg}{2m + \mu\rho C_L A R}} = \sqrt{\frac{R \times 2 \times 1.5 \times 1300 \times 9.81}{2600 + 1.5 \times 1.205 \times (0.4) \times R}} \Rightarrow V_{\text{linear}} = \frac{195.6\sqrt{R}}{\sqrt{2600 + 0.723R}}$$

The curves of these equations are plotted in Figure 4 for turn radii varying from $50m$ to $300m$. The biggest difference is that the maximum cornering speed is higher for the car with $-C_L A = 4m^2$ than the car with $-C_L A = -0.4m^2$, and that this difference increases faster and faster as the road becomes straighter. This is because, with extra downforce, the first car becomes more and more planted on the road the faster it goes, and as R increases, V_{linear} essentially approaches infinity.

However, the second car has positive lift, so it gets more unstable the faster it travels, as normal force reduces and friction reduces. Indeed, at $V_{\text{linear}} = 73.45m/s$, the lift becomes equal to the weight, and the car loses traction entirely, even on a straight road.

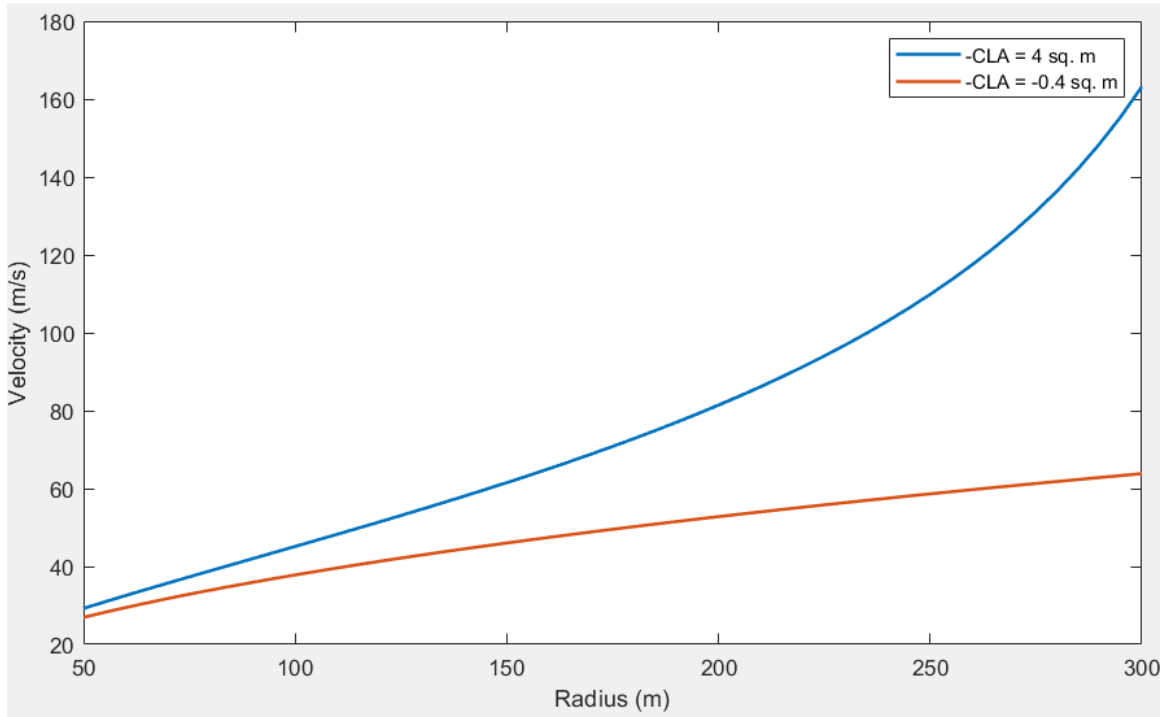


Figure 4: Variation of velocity versus curve radius for different lift coefficients

MATLAB code

Note! The code requires two .mat files - "DRIVER1_No_Aero.mat" and "DRIVER1_Aero.mat" to run properly. The .mat files have been emailed separately

Main code

```
%% MTF236 Road Vehicle Aerodynamics %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Caster Aerodynamics Assignment
%
% By -
% Ramkumar Huleppa Munavalli      huleppa@student.chalmers.se
% Anirudh Sreeram                  anirudhs@student.chalmers.se
%
% Submitted: 16 February 2025
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%% Question 1

% Answered in report

%% Question 2
% The data from the Caster driving session is taken and smoothened.
% With no Aero package:
% Load Data
data = load("DRIVER1_No_Aero.mat");
lat_acc_noAero = data.lateral_acceleration;
velocity_noAero = data.velocity;
time_noAero = data.time;

% Apply Moving Average Filter
fineness = 5;
lat_acc_smooth_noAero = movmean(lat_acc_noAero, fineness);
velocity_smooth_noAero = movmean(velocity_noAero, fineness);

% Load Data - With Aero Package
data = load("DRIVER1_Aero.mat");
lat_acc_with_Aero = data.lateral_acceleration;
velocity_with_Aero = data.velocity;
time_with_Aero = data.time;

% Apply Moving Average Filter
lat_acc_smooth_with_Aero = movmean(lat_acc_with_Aero, fineness);
velocity_smooth_with_Aero = movmean(velocity_with_Aero, fineness);

% Creating a figure, which will have two subplots
f1 = figure;

% Subplot 1: Lateral Acceleration vs Time
subplot(2,1,1);
```

```

plot(time_noAero, lat_acc_smooth_noAero, 'b', 'LineWidth', 1.5);
hold on;
plot(time_with_Aero, lat_acc_smooth_with_Aero, 'r', 'LineWidth', 1.5);

xlabel('Time (sec)');
ylabel('Lateral Acceleration (m/s2)');
title('Lateral Acceleration vs Time (Steady-State)');
legend('No Aero', 'With Aero', 'Max/Min Lat Acc (No Aero)', 'Max/Min Lat Acc (With Aero)');
grid on;
hold off;

% Subplot 2: Velocity vs Time
subplot(2,1,2);
plot(time_noAero, velocity_smooth_noAero, 'b', 'LineWidth', 1.5);
hold on;
plot(time_with_Aero, velocity_smooth_with_Aero, 'r', 'LineWidth', 1.5);

xlabel('Time (sec)');
ylabel('Velocity (m/s)');
title('Velocity vs Time (Steady-State)');
legend('No Aero', 'With Aero', 'Max/Min Vel (No Aero)', 'Max/Min Vel (With Aero)');
grid on;
hold off;

%% Given data for further questions

clear all

data.minusclamin    = -0.4;
data.minusclamax    = 6.0;
data.minusclapackage = 4.0;
data.minusclastep    = 0.1;

data.R              = 103;
data.Rmin           = 50;
data.Rmax           = 300;
data.Rstep          = 5;

data.m              = 1300;
data.v              = 46.0869;
data.mu             = 1.5;

%% Question 3

% Answered in report

%% Question 4
% Case 1 - variation of mu
i=0;
x_values = data.minusclamin:data.minusclastep:data.minusclamax;

```

```

for minuscla = data.minusclamin:data.minusclastep:data.minusclamax
    i = i+1;
    mu_syms(i) = mastereqn(minuscla, data.m, 0, data.v, data.R);
end
mu_graph = double(mu_syms);
% Presenting as graph
f2 = figure;
subplot(1,3,1);
plot(x_values, mu_graph);
xlabel('-Cl*A (sq.m)');
ylabel('Friction coefficient');
hold off

% Case 2 - variation of m
i=0;
for minuscla = data.minusclamin:data.minusclastep:data.minusclamax
    i = i+1;
    m_syms(i) = mastereqn(minuscla, 0, data.mu, data.v, data.R);
end
m_graph = double(m_syms);
% Presenting as graph
subplot(1,3,2);
plot(x_values, m_graph);
xlabel('-Cl*A (sq.m)');
ylabel('Mass (kg)');
hold off

% Case 3 - variation of v
i=0;
for minuscla = data.minusclamin:data.minusclastep:data.minusclamax
    i = i+1;
    vel_syms(i) = mastereqn(minuscla, data.m, data.mu, 0, data.R);
end
v_graph = double(vel_syms);
% Presenting as graph
subplot(1,3,3);
plot(x_values, v_graph);
xlabel('-Cl*A (sq.m)');
ylabel('Velocity (m/s)');
hold off

%% Question 5
% Calculated from a simplified formula, see report.

%% Question 6
i=0;
rad_values = data.Rmin:data.Rstep:data.Rmax;

% For -CLA = 4 sq. m.,
minuscla = 4;
for R = data.Rmin:data.Rstep:data.Rmax

```

```

        i = i+1;
        vel_6_syms_1(i) = mastereqn(minuscla, data.m, data.mu, 0, R);
    end

    i=0;
    % For -CLA = -0.4 sq. m.,
    minuscla = 0.4*(0-1);
    for R = data.Rmin:data.Rstep:data.Rmax
        i = i+1;
        vel_6_syms_2(i) = mastereqn(minuscla, data.m, data.mu, 0, R);
    end

    v_6_graph_1 = double(vel_6_syms_1);
    v_6_graph_2 = double(vel_6_syms_2);
    f3 = figure;
    plot(rad_values, v_6_graph_1, LineWidth=1.5);
    hold on
    plot(rad_values, v_6_graph_2, LineWidth=1.5);
    xlabel('Radius (m)');
    ylabel('Velocity (m/s)');
    legend("-CLA = 4 sq. m", "-CLA = -0.4 sq. m")
    hold off
    % End of main code. Ramkumar M & Anirudh S, 16.02.2025
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%% Defining function "mastereqn"

function [solution] = mastereqn(minuscla,m,mu,v,R)
syms muout mout vout
g = 9.81;
rho = 1.205;
cla = minuscla*(-1);
if m == 0
    eqn1 = (v*v)*(((2*mout)+(mu*rho*cla*R))) == (2*R*mu*mout*g);
    solution = solve(eqn1,mout);
elseif mu == 0
    eqn1 = (v*v)*((2*m)+(muout*rho*cla*R)) == (2*R*muout*m*g);
    solution = solve(eqn1,muout);
elseif v == 0
    eqn1 = (vout*vout)*((2*m)+(mu*rho*cla*R)) == (2*R*mu*m*g);
    solns = solve(eqn1,vout);
    solution = solns(2,1);
else
    eqn1 = (v*v)*((2*m)+(mu*rho*claout*R)) == (2*R*mu*m*g);
    solution = solve(eqn1,claout);
end
end

% End of code. Ramkumar M & Anirudh S, 16.02.2025
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

```

End of report.